Heyang CHEN

Nanyang Technological University and Northwestern University

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Education

Shanghai Jiao Tong University — **Shanghai, China**
Ph.D. in *Materials Science and Engineering* — Sep 2019–Mar 2025
Supervisor: Professor Xun SHI and Professor Tian-Ran WEI
Research: **thermoelectric materials** and **ductile/plastic inorganic semiconductors**

Shanghai Jiao Tong University — **Shanghai, China**
B.S. in *Materials Science and Engineering* — Sep 2015–Jun 2019

Working Experience

Nanyang Technological University — **Singapore**
Research Fellow in *Thermoelectrics*, *School of Materials Science and Engineering* — May 2025–present
Supervisor: Professor Qingyu YAN

Northwestern University — **Evanston, IL, USA**
Visiting Postdoctoral Fellow, *Department of Chemistry* — Nov 2025-present
Supervisor: Professor Mercouri G. Kanatzidis
National Academy of Sciences, *American Academy of Arts and Sciences*

Research Profile

Research Thrusts

- **2025–present | AI/LLM-enabled materials workspace:** building TEDEX – a workspace for experimental data management, transport modeling, plotting, and AI-assisted materials screening.
- **2025–present | High-entropy and new thermoelectric materials:** discovering new chalcogenides with intrinsically low thermal conductivity and probing entropy effects in semiconductor transport.
- **2019–2025 | Thermoelectric semiconductors:** composition design, synthesis, and electrical/thermal transport optimization in chalcogenide and high-entropy thermoelectric materials.
- **2019–2025 | Plastic inorganic semiconductors:** ductile silver chalcogenide and vdW semiconductors for flexible thermoelectric devices, wearable electronics, and sensors.

Selected Research Experience

- **AI/LLM and data workspace:** Developed TEDEX-oriented workflows that connect literature mining, experimental metadata, transport datasets, SPB/Debye-model fitting, and publication-quality figures, enabling reproducible reporting and AI-assisted composition screening.
- **Transport properties:** Optimized carrier/phonon transport in chalcogenide thermoelectrics; achieved near-room-temperature zT of 0.45 in high-ductility $Ag_2(S, Se, Te)$ and room-temperature zT of 0.78 in $Ag_2Se_{0.9}Te_{0.1}$.
- **Mechanical properties:** Established composition–structure–property relations across silver chalcogenides, linking symmetry, defects, and texture to plasticity; demonstrated tensile strain up to 97%, bending strain of 15%, compression strain of 50%, and fracture toughness of 4.5 MPa $m^{1/2}$.
- **Applications and devices:** Fabricated substrate-free in-plane thermoelectric devices with 13.6 mV open-circuit voltage and 12.8 μW output under a 30 K temperature difference. Prepared plastic p- and n-type $Ag_2(S, Se, Te)$ fibers and developed high-precision thermocouples with temperature-measurement accuracy two orders of magnitude higher than commercial thermocouples.

Experimental, Computational, and AI Skills

- **Synthesis:** Bridgman and CVT crystal growth for 2D vdW materials; vacuum melting, energy ball milling, and spark plasma sintering for thermoelectric semiconductors.
- **Characterization and testing:** XRD/JADE/VESTA/GMS3; SEM/TEM; Seebeck, electrical, and thermal transport; magnetic/mechanical testing; self-built electrical-transport setups.
- **Physical modeling and computation:** Debye-model analysis, single parabolic band fitting, DFT calculations for semiconductor band structures, phonons, and Boltzmann transport.
- **AI-for-materials workflows:** Working knowledge of LLM fundamentals, including transformers, embeddings, RAG, prompt/workflow design, and model evaluation; studied materials-AI pipelines for data curation, featurization, graph/neural representations, surrogate modeling, active learning, and high-throughput screening.
- **Programming and tools:** Python/Matplotlib, Git/GitHub, OpenAI Codex, OriginLab, Office 365, LaTeX, Adobe Photoshop, and Adobe Illustrator.
- **Language:** Fluent in academic written and spoken English.

Selected Publications

1. **Heyang Chen**, Chenlu Shao, Shaoji Huang, Zhiqiang Gao, Haoran Huang, Zhenyu Pan, Kunpeng Zhao, Pengfei Qiu, Tian-Ran Wei, Xun Shi, "High-entropy cubic pseudo-ternary $Ag_2(S,Se,Te)$ materials with excellent ductility and thermoelectric performance," *Advanced Energy Materials*, 2024, 14, 2303473.
2. **Heyang Chen**, Tian-Ran Wei, Kunpeng Zhao, Pengfei Qiu, Lidong Chen, Jian He, Xun Shi, "Room-temperature plastic inorganic semiconductors for flexible and deformable electronics," *InfoMat*, 2021, 3, 22–35.
3. **Heyang Chen**, Tian-Ran Wei, Xun Shi, "Plastically deformable inorganic semiconductors," *Materials Lab*, 2023, 2, 230008.
4. **Heyang Chen**[#], Shenghong Ren[#], Hengyang Feng, Tian-Ran Wei, Ling Fu, Zhenyu Pan, Kunpeng Zhao, Xiuyan Li, Xun Shi, "Robust mechanical properties of $Ag_2Se_{1-x}Te_x$ thermoelectric materials," *National Science Open*, 2025, 4, 2025005. [#]Equal contribution.
5. Zixun Zhang, **Heyang Chen**, Tian-Ran Wei, Kunpeng Zhao, Xun Shi, "Data-driven discovery of high-performance multicomponent solid solution thermoelectric materials," *Materials Today Energy*, 2022, 28, 101070.
6. Shenghong Ren, **Heyang Chen**, Huangliu Fu, Haoran Huang, Tian-Ran Wei, Xun Shi, Xiuyan Li, "The structural origin of extraordinary plasticity in polycrystalline semiconductors with low symmetry," *Science Advances*, 2025, 11, eadu9205.
7. Zixun Zhang, Kunpeng Zhao, **Heyang Chen**, Qingyong Ren, Zhongmou Yue, Tian-Ran Wei, Pengfei Qiu, Lidong Chen, Xun Shi, "Entropy engineering induced exceptional thermoelectric and mechanical performances in $Cu_{2-y}Ag_yTe_{1-2x}S_xSe_x$," *Acta Materialia*, 2022, 224, 117515.
8. Haotian Gao, Kunpeng Zhao, Hexige Wuliji, Min Zhu, Beibei Xu, He Lin, Liting Fei, Hongyao Zhang, Zhengyang Zhou, Jingdan Lei, **Heyang Chen**, Shun Wan, Tian-Ran Wei, Xun Shi, "Adaptable sublattice stabilized high-entropy materials with superior thermoelectric performance," *Energy & Environmental Science*, 2023, 16, 6046–6057.

Full publication list available on Google Scholar.

Selected Presentations

- MRS Spring Meeting & Exhibit, Honolulu, USA, 2026, Oral Presentation.
- ICMAT, Singapore, 2023, Oral Presentation.
- International Conference on High-performance Ceramics, Suzhou, China, 2022, Oral Presentation
- Chinese Conference on Thermoelectrics, Tongxiang, China, 2023, Poster.